

Scene-aware Motion Redirection in Telecommunication

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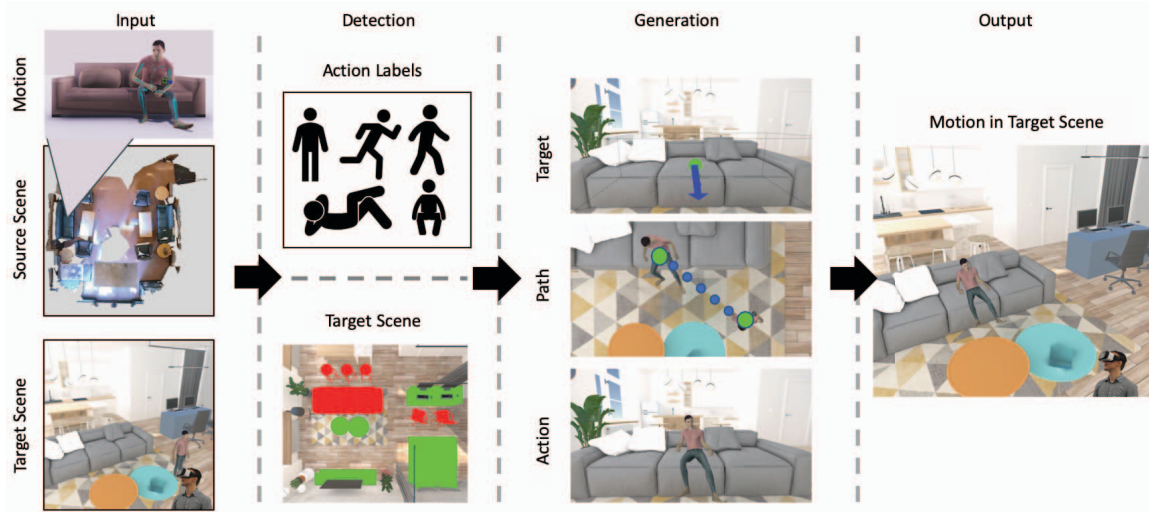


Figure 1: We propose a pipeline to redirect the motion of user A in a source scene to a target scene where user B is. With the mapping, user A's motion is natural and reasonable to present in the target scene for user B to interact with and observe. Given a source scene, a target scene, and a motion sequence in the source scene, we apply algorithms to recognize the motion and the scene's semantics. Then, according to the semantics of the target scene, we redirect the input motion into an target position, a moving path, and an action to form a directed motion presented in the target scene.

ABSTRACT

We propose a motion redirection system for virtual reality and augmented reality, automatically detecting a person's action in a source scene and redirecting it to a new target scene. The redirected motion is augmented in the target scene based on the understanding of the target scene's semantics and layout. The results show that the system may facilitate telecommunication across scenes, improving user interaction experiences.

Index Terms: Human-centered computing—Visualization—Visualization techniques

1 INTRODUCTION

Nowadays, telecommunication has become one of the most popular means for people to socialize. Different from traditional communication ways such as voice, words, and streaming, emerging techniques of VR and AR devices allow users to telecommunication in a vivid and immersive way. However, the task of visualizing the users' motions across scenes and improving the online communication experience remains challenging.

For the situation of the source scene and the target scene are the same, replaying the motion is a plausible solution [5]. But for different spaces, the problem of action alignment needs to be solved. Some studies focus on the correspondence among the locations of different spatial motion [1], contact [4], and so on. However,

the task of spatial-level motion redirection receives less attention. Therefore, this paper proposes an action-generation method that can use scene information and actions in the source space to generate corresponding actions in the target scene.

2 SYSTEM FRAMEWORK

We aim to generate the same action that suits the target scene. Figure 1 shows the framework of the approach. With a wearable VR helmet, the scene and the user's actions are detected. Our system takes user actions and interactive objects as input and generates synchronously vivid actions in the output target scene. By reverse deployment, bidirectional action generation can be achieved. The method consists of two components: detection and generation.

Detection. In the detection part, there are two modules: action detection to obtain the label of the user's action and scene detection to obtain the bounding box of the scene's objects.

Action Detection. We analyzed and designed some action tags for the action processing module. The user's action label is obtained through behavior analysis using the spatial pose relationship.

We obtained the training data by three steps: participants wore a headset to perform the desired actions; we extracted three features from the collected data to train the regressor, i.e., position, height, and velocity. After data labeling, a three-layer fully connected neural network is trained to learn the relationship between the action label and a feature vector encoding action attributes.

Scene Detection. For the scene detection module, according to the application type, we can take different ways to obtain the detection results. In a Virtual Reality scene, we can retrieve the 3D model through the model's ID, obtaining the occupation of each model in the virtual scene. If a Mixed Reality scene, we first scan the scene through the HoloLens headset to obtain the 3D model of the scene. Secondly, the scene is detected by RfD-Net [3], after object

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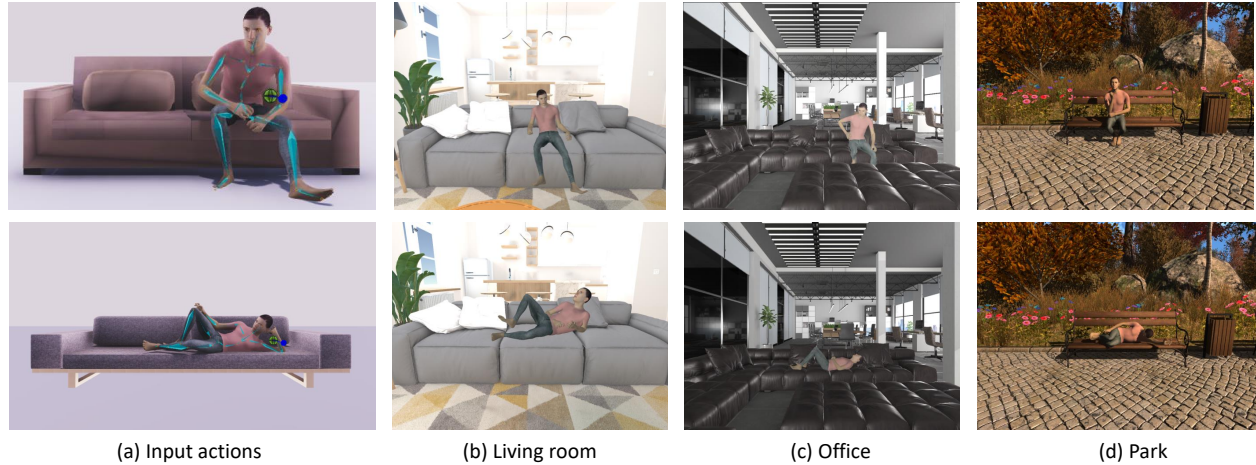


Figure 2: Result of some generation examples. (a) Input actions. (b-d) generated actions in different scenes.

detection, space transformation, and shape generation to obtain a relatively complete interactive object model. Finally, the model is voxelized to generate interactive object models.

For N point cloud input, we take the 3D coordinates of each point as input and get $P \in \mathbb{R}^{N \times 3}$. We use VoteNet as the basis of the object detection network to generate N_p candidate boxes. The candidate box is expressed as a box parameter of D_b dimension, including the box center $c \in \mathbb{R}^3$, and the box scale $s \in \mathbb{R}^3$, box orientation angle $\theta \in \mathbb{R}$, semantic label l , target score s_{obj} . Use a 2-layer multilayer perceptron for regression training. After obtaining N_d positive candidate boxes, the point cloud data needs to be processed. Firstly, the point cloud is grouped according to the candidate boxes. Then, a 3D shape is represented by an occupancy function.

Generation. Our method includes three modules in the generation part: target generation, path planning, and action generation. Based on the detection results in the previous component, our system first generates a target object with the corresponding interaction position and orientation for the redirected avatar. Second, our system finds a possible path from the avatar's current location to the target location. Finally, an action sequence is generated from the current location to the target location.

Target Generation. When obtaining an interactive object, we need to learn how to interact with the object based on the geometric information of the object. For example, given a chair, we know that we should sit on the top, facing forward, rather than sitting on the back or facing the back. Therefore, referring to [2], we design an object prediction network for learning object interaction orientation.

To generate multiple plausible positions and orientations, we choose cVAE as the base model. The encoder uses the object's $g^p \in \mathbb{R}^3$, $g^d \in \mathbb{R}^3$ position, orientation, target object as input, output hidden variable Z_{target} . The decoder takes hidden variables and objects as input and outputs the predicted position towards $\hat{g}^p, \hat{g}^d$.

Path Planning. We use the A* algorithm to add the ability to avoid obstacles. When a target is selected, the avatar first calculates a path to the target. The avatar starts acting when reaching a place close to the target.

Action Generation. We use cVAE as the basic framework for action generation. The character's actions are generated based on the character's previous state and interacting objects. The whole system includes two processes: encoding and decoding. The encoding process encodes the previous state, current state, and interactive objects into hidden variables Z . The decoding process takes hidden variables, the previous state, and interactive objects as input to predict the next state of the character. For final use, we only use the decoder to predict actions.

3 EXPERIMENT

We implemented our method using C# and Unity 2019 and ran on a PC equipped with 32GB of RAM, an Nvidia 2070s graphics card with 8GB of memory, and a 3.0GHz Intel i7-9700K processor.

To validate the effectiveness of our method, we experiment on three scenes: a living room, an office, and a park. We test two tasks, sitting and lying down in each of them.

Fig. 2 shows the generated results. For the various input actions, the output is quite different. Comparing sitting and lying down, we can see that the generated actions are divided into two categories. For the diverse scene, the output changes slightly. The avatar's pelvis and the furniture's surface are the same height. The action generation considers furniture shapes so that the generated action fits the model.

The experiment results show that our approach can be applied to different scenes. Our approach learn the action label and interaction object message by detecting the environment structure. Then, our approach can generate a more reasonable action for the avatar.

4 CONCLUSION

In this paper, we introduce an approach for the user action redirection in a telecommunication scenario, helping users improve the interaction experiences. The action generated by our approach could meet the requirements of both user preference and spatial rationality. Results show that our method can be easily applied in different scenes.

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